Formalization and Interaction: Toward a Comprehensive History of Technology-Related Knowledge in Early Modern Europe

Marcus Popplow, *Technische Universität Berlin*

Abstract: Recent critical approaches to what has conventionally been described as "scientific" and "technical" knowledge in early modern Europe have provided a wealth of new insights. So far, the various analytical concepts suggested by these studies have not yet been comprehensively discussed. The present essay argues that such comprehensive approaches might prove of special value for long-term and cross-cultural reflections on technology-related knowledge. As heuristic tools, the notions of "formalization" and "interaction" are proposed as part of alternative narratives to those highlighting the emergence of "science" as the most relevant development for technology-related knowledge in early modern Europe.

Considerable advancements have been made in recent decades in the study of technology-related knowledge in early modern Europe. Stimulating contributions have been provided by researchers of various disciplinary backgrounds—most prominently not by historians of technology, but by historians of science. They have developed versatile methodological tools that have proven useful for investigations of historical transformations of knowledge. Drawing from the insights gained in the visual, the material, and the spatial turn of the history of science, these scholars have stressed the importance of "practical" knowledge for the generation of "scientific" insights, while being careful to avoid an anachronistic reading of modern concepts of "science" into early modern sources. This study of technology-related knowledge has also been fruitfully addressed from the perspective of a social history of knowledge. Notions of "experts" and "expertise," usually employed for twentieth-century developments, have convincingly been extended to early modern actors, who often cannot adequately be defined as "scientists" or "technicians." In that context, the relevance of administrative expertise has been stressed as well. Based on a different interest, economic historians have argued that transformations in technical knowledge produced relevant eco-

Fachgebiet Technikgeschichte, Technische Universität Berlin, Sekr. H 67, Institut für Philosophie, Literatur-, Wissenschafts- und Technikgeschichte, Straße des 17. Juni 135, D-10623 Berlin, Germany; m.popplow@tu-berlin.de. I am very grateful to Karel Davids for organizing this *Isis* Focus section; he and the other participants, as well as an anonymous reviewer, provided instructive comments on preliminary versions of this text.

Isis, volume 106, number 4. © 2015 by The History of Science Society.
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nomic effects, particularly during the early stage of British industrialization. Historians studying early modern artisans, in turn, have extended a tradition of investigating processes of knowledge circulation among them by also addressing more systematically methodological approaches to artisanal knowledge. They stress the complexity of this knowledge and argue that artisans drew from many more types than the usual restricted sense of just possessing "implicit" or embodied knowledge would suggest.[1] As will be explained in more detail below, all these different strains of research use different methodological concepts to summarize their particular approach to what has long been called the intersection of "scientific" and "technical" knowledge in the early modern period.

The state of the art in this area is thus as stimulating as it is somewhat confusing due to the lack of systematic discussions of these various methodological approaches. While nearly all these studies are convincing in their own right, this essay asks for possible steps toward a more comprehensive evaluation of transformations of technology-related knowledge in early modern Europe.

In the first section, such transformations are briefly sketched, paying particular attention to long-term developments since the Middle Ages. In the second section, the concepts mentioned above with which these transformations have recently been analyzed are briefly surveyed in more detail. The third section argues that notions of the "formalization" of technology-related knowledge and the subsequent "interaction" of corresponding knowledge practices are particularly suited to analyze transformations of this knowledge in early modern Europe.

"Technology" in this essay is used in the now common double meaning not used by early modern actors; this use only evolved in the nineteenth and twentieth centuries. Since then, the term has denoted technical artifacts *and* skills.[2] At the same time, speaking of "technology-related knowledge" serves to highlight additional aspects of its transformation in the early modern period. These include the knowledge necessary for devising and manufacturing

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[1] Among numerous case studies see those in the history of science with explicit methodological reflections. These include, for example, Ursula Klein, ""Technoscience avant la lettre,"" *Perspectives on Science*, 2005, *13*:227–266; Lissa Roberts, Simon Schaffer, and Peter Dear, eds., *The Mindful Hand: Inquiry and Invention from the Late Renaissance to Early Industrialisation* (Chicago: Univ. Chicago Press, 2008); Pamela H. Smith, *The Body of the Artisan: Art and Experience in the Scientific Revolution* (Chicago: Univ. Chicago Press, 2004); and Smith and Benjamin Schmidt, eds., *Making Knowledge in Early Modern Europe: Practices, Objects, and Texts, 1400–1800* (Chicago: Univ. Chicago Press, 2007). Contributions by historians of technology comprise Pamela O. Long, *Artisan/Practitioners and the Rise of the New Sciences, 1400–1600* (Corvallis: Oregon State Univ. Press, 2011); and Hélène Vérin and Pascal Dubourg Glatigny, eds., *Réduire en art: La technologie de la Renaissance aux Lumières* (Paris: Éditions de la Maison des Sciences de l'Homme, 2008). From the perspective of a social history of knowledge and on the relevance of administrative expertise see *Expertise: Practical Knowledge and the Early Modern State*, ed. Eric Ash, *Osiris*, 2010, N.S., *25*. On economic historians' points of view see Joel Mokyr, *The Enlightened Economy: An Economic History of Britain, 1700–1850* (New Haven, Conn.: Yale Univ. Press, 2009); and Margaret C. Jacob, *The First Knowledge Economy: Human Capital and the European Economy, 1750–1850* (Cambridge: Cambridge Univ. Press, 2014). For historians investigating knowledge circulation among artisans see, for example, Liliane Hilaire-Perez and Catherine Verna, "Dissemination of Technical Knowledge in the Middle Ages and the Early Modern Era," *Technology and Culture*, 2006, *47*:536–565. On methodological approaches to understanding artisanal knowledge see, for example, Stephen R. Epstein and Maarten Prak, *Guilds, Innovation, and the European Economy, 1400–1800* (Cambridge: Cambridge Univ. Press, 2008); Bert de Munck, *Technologies of Learning: Apprenticeship in Antwerp Guilds from the 15th Century to the End of the Ancien Régime* (Turnhout: Brepols, 2007); and Sven Dupré and Christoph Lüthy, eds., *Silent Messengers: The Circulation of Material Objects of Knowledge in the Early Modern Low Countries* (Berlin: LIT Verlag, 2011).

[2] See Erik Schatzberg, "*Technik* Comes to America: Changing Meanings of *Technology* before 1930," *Technology and Culture*, 2006, *47*:486–512.

technical objects, and the knowledge that users or writers employed to interact with such technical objects and to reflect on the objects and skills.

The notions of "formalization" and "interaction" are introduced here as heuristic tools. They are deliberately intended to offer an alternative narrative to that of underlining "science" as the most relevant development of technology-related knowledge in early modern times.[3] As many studies have convincingly argued, despite the frequent and complex efforts to address technological problems with advanced mathematical knowledge, even the technological basis of early industrialization was, in fact, hardly ever generated or decisively shaped by the "application of science" in such a sense.

TRANSFORMATIONS OF TECHNOLOGY-RELATED KNOWLEDGE IN EARLY MODERN EUROPE

While devising and manufacturing technology in early modern Europe essentially remained dependent on embodied craft skills, this period saw the emergence of a number of new types of technology-related knowledge. This becomes especially clear from a long-term view that takes into account the transformations of the contexts in which technology-related knowledge was generated, stored, and communicated during the Middle Ages *and* the early modern period. Around 1800, this panorama was much more diversified than had been the case around 1400. This does not mean that artisans and other technical experts had become cleverer, but that new frameworks for certain types of technical knowledge had been established.[4]

Modern historical research has analyzed many of such rather loosely connected knowledge practices that certain actors increasingly made use of in producing and using technology in early modern Europe. Such knowledge was stored, developed, and communicated in representational media such as technical drawings and treatises, and it was collected, discussed, and taught in institutions, including scientific academies, economic societies, and engineering schools. Related processes also concerned practices to employ particular objects such as scaled-down models and measuring instruments; to conduct field trials; to engage in learned correspondence; and to set up privileges for inventions, patents, or prize contests.[5]

Compared to embodied knowledge, it would be wrong from the start to attach the label of "progress" to these early modern transformations. Despite abundant case studies, it is still unclear what definite effects they had on the history of early modern technology. In the Middle Ages, as in earlier periods, personal experience and instruction obviously were extremely powerful tools for the conception and transmission of new technologies—even in regard to complex processes like the structural engineering of Gothic cathedrals or procedures

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[3] Most recently, Margaret C. Jacob, although in fact delineating a much broader picture of eighteenth- and nineteenth-century technical knowledge, on the conceptual level surprisingly insists that "applied scientific learning," and in particular "mechanical knowledge, derived largely but not exclusively from Newtonian science," was of crucial importance for the "Industrial Revolution." See Jacob, *Knowledge Economy* (cit. n. 1), pp. 223 and 1.

[4] Despite a wealth of case studies, there is as yet no comprehensive, in-depth analysis summarizing the history of early modern technology with an emphasis on these issues of knowledge transformation.

[5] Joel Mokyr has surveyed such developments comprehensively in his essay "The Intellectual Origins of Modern Economic Growth," *Journal of Economic History*, 2005, 65:286–351. For Britain, Celina Fox has described these developments in great detail in *The Arts of Industry in the Age of Enlightenment* (New Haven, Conn.: Yale Univ. Press, 2009). In regard to the representational media mentioned, Mokyr has labeled them "storage devices" for technical knowledge; see "The Intellectual Origins," p. 300. For analyzing the role of such representational media in the modern period, different concepts such as Bruno Latour's "immutable mobiles" have been proposed.

in mining and smelting. Many arts and crafts spread by way of the migration of individuals over large distances, a process often actively promoted by guilds, town councils, and territorial powers. Such embodied knowledge, shaped by experience and instruction, continued to be of essential importance in the early modern period as well. The establishment of additional ways of generating, storing, and communicating technology-related knowledge, however, marks a quite distinctive difference between the Middle Ages and the early modern period.[6]

Knowledge related to the employment of such representational media was often concentrated in certain locations, in particular at court, but also in the private homes of technical experts. One might thus speak of "clusters" in which such diverse knowledge traditions were continuously confronted with each other.[7]

It goes without saying that only selected bodies of information could be stored in and communicated by such representational media as drawings, models, and treatises. At the same time, considerable portions of embodied artisanal knowledge assembled over years or decades of experience could not be transferred into these media for subsequent replication in practice. Thus, the employment of media resulted more often in generating new types of knowledge. Access to these kinds of knowledge was limited by diverse mechanisms, including access to the (very exclusive) institutions mentioned, or access to technical drawings and technical literature that required corresponding skills. Connected to these limitations of access were new mechanisms for dealing with "secret" and "public" knowledge that were much more complex than these opposing categories might suggest at first sight.[8]

The processes briefly sketched here were not limited to the spheres of devising and manufacturing technical artifacts, which usually are at the focus of studies of technology-related knowledge in the early modern period. Without going into further detail here, these processes generated knowledge related to the use and consumption of technologies, as well as such knowledge related to reflection on technical objects and skills. Examples include discourses on the benefit of the arts and crafts for the common well-being and on the legitimacy of consumption patterns. Technology only becomes effective in society when it is produced *and* used—and both these spheres, in the early modern period, were affected by the transformation of technology-related knowledge.

METHODOLOGICAL APPROACHES TO EARLY MODERN TECHNOLOGY-RELATED KNOWLEDGE

As already mentioned above, owing to the many studies published in the last two decades there is surely no shortage of concepts for interpreting all such developments of technology-

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[6] The developments referred to in this essay, which many case studies discuss, have been briefly summarized for a series of entries in the *Enzyklopädie der Neuzeit*, covering the period 1450–1850; an English translation is in preparation. See in *Enzyklopädie der Neuzeit*, ed. Friedrich Jäger, 15 vols. (Stuttgart: Metzler Verlag, 2004–2012), Marcus Popplow, "Popularisierung, Technik," Vol. 10 (2009), cols. 200–204; Popplow, "Technische Literatur," Vol. 13 (2011), cols. 292–297; Popplow, "Technisches Modell," Vol. 13 (2011), cols. 298–303; Popplow, "Technisches Zeichnen," Vol. 13 (2011), cols. 303–310; Bernd Hausberger, Popplow, and Reinhold Reith, "Technologietransfer," Vol. 13 (2011), cols. 319–334; Popplow and Reith, "Wandel, technischer," Vol. 14 (2011), cols. 594–614; and Popplow and Reith, "Umwelt und Technischer Wandel," Vol. 15 (2012), cols. 1075–1090.

[7] This idea of "clusters" in many respects surely corresponds to that of Pamela O. Long's "trading zones," discussed in this Focus section. While Long's focus is on contacts between artisans and scholars, such "clusters," in a more general sense, would focus on the coming together of different types of knowledge and related practices.

[8] For the study of such processes see, for example, Pamela O. Long, *Openness, Secrecy, Authorship: Technical Arts and the Culture of Knowledge from Antiquity to the Renaissance* (Baltimore: Johns Hopkins Univ. Press, 2001); and "Openness and Secrecy in Early Modern Science," ed. Karel Davids, special issue, *Early Science and Medicine*, 2000, 9.

related knowledge in early modern Europe. Some of this research primarily argues for new conceptual approaches to overcome such binary oppositions as "science" versus "technology" and "theory" versus "practice"; Ursula Klein has thus proposed to extend the notion of "technoscience," usually employed for twentieth-century developments only, to the early modern period. In this Focus section, Pamela O. Long draws attention to "trading zones" in which different kinds of "practical" and "theoretical" knowledge were exchanged. Lissa Roberts, Simon Schaffer, and Peter Dear have proposed the "mindful hand" and "inventive intersections" as concepts denoting the continuous overlapping of these fields of "theory" and "practice." It is also they who have perhaps most explicitly argued for avoiding, wherever possible, the notions of "science" and "technology" in early modern contexts, as they can be anachronistic and misleading.[9]

Other approaches have focused on analyzing the increasing use of representational media for storing and communicating technology-related knowledge in the early modern period; this is traditionally referred to as the "codification" of knowledge. While also investigating in detail the "embodied" knowledge of highly specialized artisans, Pamela Smith has investigated this employment of media for the sixteenth and seventeenth centuries. Ursula Klein has studied comparable topics in different contexts, namely in eighteenth-century porcelain manufactories. In that context, she used the notion of "depersonalization" to describe a central aspect of the differences between "embodied" knowledge and knowledge stored and communicated first and foremost in written form. Hélène Vérin and Pascal Dubourg Glatigny have drawn attention to the concept of "réduire en art," which some authors of early modern technical treatises used in attempts to reduce the complexity of problems with which they were confronted by turning them into general rules.[10]

Joel Mokyr has set up a quite different distinction between two modes of knowledge that he calls prescriptive and propositional. In particular, his distinction is intended to highlight transformations that technology-related knowledge is subjected to when generated with the aim of economic application. More recently, however, Mokyr and a number of other scholars have increasingly used the more comprehensive concept of "useful knowledge."[11] Employed as an analytical tool for historical research, such "useful knowledge" is usually investigated with regard to the developments outlined in the preceding section.

Eric Ash's proposal to apply the notions of "experts" and "expertise" to early modern technology-related knowledge is situated on a quite different and all the more interesting level, insofar as it draws special attention to the social and administrative contexts in which such knowledge was contested and debated. His focus is thus not so much on the content and formal structure of this knowledge but on the procedures with which it was applied.[12]

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[9] Klein, "Technoscience avant la lettre" (cit. n. 1); Long, *Artisan/Practitioners* (cit. n. 1); and Roberts, Schaffer, and Dear, *Mindful Hand* (cit. n. 1).

[10] For an example of this work see Pamela H. Smith, "Why Write a Book? From Lived Experience to the Written Word in Early Modern Europe," *Bulletin of the German Historical Institute, Washington*, 2010, 47:25–50; Ursula Klein, "Depersonalizing the Arcanum," *Technology and Culture*, 2014, 55:591–621; and Vérin and Dubourg Glatigny, *Réduire en art* (cit. n. 1).

[11] On prescriptive and propositional knowledge see Joel Mokyr, *The Gifts of Athena: Historical Origins of the Knowledge Economy* (Princeton, N.J.: Princeton Univ. Press, 2002). On "useful knowledge" see Mokyr, "Intellectual Origins" (cit. n. 5); "Special Issue: Reflections on Joel Mokyr's *The Gifts of Athena*," ed. Maxine Berg, *History of Science*, 2007, *45*, no. 2; "Conceptualising the Production and Diffusion of Useful and Reliable Knowledge in Early Modern Europe," ed. Ian Inkster, *History of Technology* 2012, *31*; and Patrick O'Brien, "Historical Foundations for a Global Perspective on the Emergence of a Western European Regime for the Discovery, Development, and Diffusion of Useful and Reliable Knowledge," *Journal of Global History*, 2013, 8:1–24.

[12] Eric Ash, "Introduction: Expertise and the Early Modern State," *Osiris*, 2010, 25:1–24. Lothar Schilling and Regina Dauser

All these different suggestions to conceptually address early modern technology-related knowledge, with their respective features and applicability, have only seldom been comprehensively discussed and synthesized. This might also be one of the reasons why the complexity of the discussions that have taken place has not yet been integrated into comprehensive studies on early modern technology. These debates have been perceived even less often outside the small community of historians of early modern technology and science. For example, the debate on the "Great Divergence" has so far hardly taken note of issues concerning technology-related knowledge. In contrast, its participants still quite often reiterate outdated dichotomies of "practical" and "scientific" knowledge when they raise any respective questions.[13]

"FORMALIZATION" AND "INTERACTION" OF TECHNOLOGY-RELATED KNOWLEDGE

A plurality of methodological approaches to a historical topic is surely no problem in itself, especially when it stimulates discussion within a discipline or among neighboring ones. This is also true of conceptualizing early modern technology-related knowledge, as discussed here. However, no doubt there are situations in which a more comprehensive approach might be helpful. This is particularly the case when research results have to be communicated to other historical and scientific disciplines or to the interested public.

What requirements would such a comprehensive approach to technology-related knowledge in early modern Europe ideally have to fulfill? One might summarize these as follows. To begin with, corresponding concepts should be devised to broadly address technology-related transformations of knowledge peculiar to the early modern period, as sketched above, and to comprise knowledge required for producing, using, and reflecting on technology. Secondly, to avoid reiterating the anachronistic dichotomy of "scientific" versus "technical" knowledge, concepts should reflect the recent methodological refinements sketched in the preceding section, and be applicable to the wealth of case studies published in this field. Thirdly, even if on many occasions it is helpful to distinguish embodied and codified knowledge on the conceptual level, such distinctions should neither privilege the investigation of codified knowledge, nor should they presuppose its higher relevance compared to embodied knowledge. They should instead leave ample opportunities for discussing the interaction of both knowledge traditions with which historical actors were continuously confronted. Finally, this comprehensive approach should not be based on Eurocentric categories, but should also allow fine-grained cross-cultural comparisons.

The several approaches summarized in the preceding section do not directly match all these requirements; perhaps with the exception of the concept of "useful knowledge," they have been conceived with different interests than the aim of comprehensively covering transformations of technology-related knowledge. Many of them, even if implicitly, focus on investigating processes of the codification of knowledge. There surely are good reasons for employing the notion of codification to address, in particular, the rapidly increasing

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have proposed a comprehensive approach to early modern "state-relevant knowledge" (Savoirs d'État) that, among other things, also includes technical expertise. See Schilling and Dauser, *Grenzen und Kontaktzonen: Rekonfigurationen von Wissensräumen zwischen Frankreich und den deutschen Ländern 1700–1850*, and *Discussions*, ed. Schilling and Dauser, 2002, 7 (www.perspectivia.net/content/publikationen/discussions/7-2012).

[13] As an exception see Karel Davids, *Religion, Technology, and the Great and Little Divergences: China and Europe Compared, c. 700–1800* (Leiden: Brill, 2013).

expression of technical knowledge in written form in the early modern period. However, codification was only one element of early modern transformations of technology-related knowledge. These transfomations also comprised the establishment of procedures for which "codification" presupposes too high a degree of explicit, written systematization of knowledge, and thus does not seem adequate. This could be in regard to dealing with scaled-down models or measuring instruments, the informal sharing of knowledge by means of learned correspondence, or the establishment of conventions of technical drawing.

An emphasis on codification is very understandable from the point of view of a broadly conceived history of science. However, from the point of view of the history of early modern technology, one must not forget that in any technological sector, the realization of technical processes still primarily depended on the application of embodied knowledge. This continued to be essential in the arts and crafts and in large-scale engineering processes, and was of equally crucial relevance to the application of media-based forms of knowledge. As has been stressed quite frequently, it is very important to create ample opportunities to discuss embodied and other forms of artisanal knowledge with regard to early modern developments.[14]

The concept of "useful knowledge," among those considered above, covers the widest variety of technology-related transformations of knowledge in the early modern period and has also stimulated the most extensive and fruitful discussions in recent research.[15] However, as an analytical tool, the notion of "useful knowledge" always poses the problem of demarcating the border between "useful," and "useless" or "futile," knowledge. Even when used by early modern authors, the notion of "usefulness" was frequently employed with strategic intentions rather than as a neutral term. It often quite deliberately served to conceal particular interests of those who would have most directly drawn profit from projects communicated as "useful" to a broader public. In recent research, this aspect of defining the category of "usefulness" has been avoided somewhat. It has usually been understood either, in a general sense, as knowledge applicable to all sorts of technical processes or, more specifically, as contributing toward the generation of economic growth. However, the functions that transformations of technology-related knowledge indeed had in various early modern contexts were hardly limited to the creation of economic growth. Despite the fruitful discussions it continues to generate, there might thus be good reasons for abstaining from using the term "useful knowledge" when the aim is a comprehensive highlighting of knowledge transformations.

In light of these limitations of the concepts of "codified" and "useful" knowledge to comprehensively conceptualize transformations of technology-related knowledge, the proposal put forward here is explicitly intended to allow a joint investigation of embodied and codified knowledge, and also include cross-cultural comparisons. The notions of "formalization" and "interaction" are proposed as a very basic means to serve these functions, while also covering the growing number of types of technology-related knowledge that appear as a particular characteristic of early modern Europe as compared to earlier periods.[16]

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[14] Maxine Berg, "The Genesis of 'Useful Knowledge,'" in Berg, *Reflections* (cit. n. 11), pp. 123–133; and Liliane Hilaire-Perez, "Technology as a Public Culture in the Eighteenth Century: The Artisans' Legacy," *ibid.*, pp. 135–153.

[15] See the titles cited in note 11.

[16] On other occasions, I have discussed a development that occurred as part of these new processes of formalization of technology-related knowledge in the early modern period, but it is not covered in detail in this essay. This is the emergence of "cultures of innovation"—that is, contexts in which technical innovation was explicitly promoted as a desirable aim in early modern Europe. These cultures of innovation were not necessarily directed toward producing spectacular inventions but much rather aiming at, in modern words, incremental innovations and extended diffusion of already existing artifacts and products in a given place or territory. Part of the increasing complexity created by the interaction of the formalized types of knowledge

The notion of "formalization" is intended to refer to any sort of knowledge and the practices with which it was exchanged. It not only covers knowledge related to representational media like texts or drawings, and to formal institutional settings for technical research and education; formalization also concerns embodied knowledge—for example, with regard to the acquisition of a skill in a year-long or decade-long process of practicing a craft by handling tools and materials. Also, the social settings in which arts and crafts were practiced since ancient times were characterized by formalization of knowledge—for example, by means of rules imposed by guilds for various stages of apprenticeship. From this perspective, the increasing use of writing and drawing in late medieval and early modern technology just appears as an additional, albeit quite different, aspect of formalization of technology-related knowledge. Conceived in this way, processes of formalization were thus neither alien to time periods preceding the early modern period, nor were they exclusively a European phenomenon. Simply by bringing together highly qualified artisans, any ancient building site from the pyramids to Angkor saw the interaction of various traditions of formalized knowledge represented by artisans with different specializations and of different origins. And formalization related to representational media can be perceived in different regions of the world, such as the Ancient Near East, later the Ottoman Empire, and China in particular. Taking various aspects of the formalization of technology-related knowledge as a yardstick for describing its appearance at certain locations and in certain economic, social, and political contexts might thus prove reasonably acceptable for more long-term as well as for cross-cultural comparisons.

Such processes of formalization should, in a second step, be analyzed with regard to the continuous "interaction," the second key term used here, of various types of technology-related knowledge and the practices related to them. This notion adopts the procedural approach so convincingly advocated by Eric Ash. The emphasis on interaction is of particular relevance for the long-term perspective taken here, because such processes of interaction took on new dimensions when various types of technology-related knowledge and corresponding practices multiplied at certain locations, and also increased for certain individuals during the early modern period. With regard to the spatial agglomeration of formalized knowledge in certain places—for example, courts, academies, or urban contexts—one might thus speak of the previously mentioned "clusters" where various sorts of formalized knowledge opened up diverse possibilities for historical actors to relate them to each other. To be sure, the result of this agglomeration was not homogeneity, but confusion and contradiction. What writers proposed or mathematically deduced in, for example, technological treatises often did not prove realizable in practice or did not correspond to formalized experience in the arts and crafts. As is well known, the material qualities of certain objects in systematic experimentation did not necessarily correspond to expectations based on rules previously laid down in written form. In each of the branches of technology in which these additional types of formalized knowledge emerged during the early modern period, one can thus perceive complex attempts to homogenize various approaches to the same technical problems; these attempts usually created even more contradictions that, again, needed clarification. With regard to the actual

discussed above was that the generation of technical novelties, which have actually been produced ever since, was increasingly and explicitly established as a desirable goal; and this was a new phenomenon compared to innovation processes in preceding periods like the Middle Ages. See Marcus Popplow, "Die Ökonomische Aufklärung als Innovationskultur des 18. Jahrhunderts zur optimierten Nutzung natürlicher Ressourcen," in *Landschaften agrarisch-ökonomischen Wissens: Strategien innovativer Ressourcennutzung in Zeitschriften und Sozietäten des 18. Jahrhunderts*, ed. Popplow (Münster: Waxmann Verlag, 2010), pp. 2–48; and Popplow, "Knowledge Management to Exploit Agrarian Resources as Part of Late-Eighteenth-Century Cultures of Innovation: Friedrich Casimir Medicus and Franz von Paula Schrank," *Annals of Science*, 2012, 69:413–433.

practices connected to various forms of technology-related knowledge, the notions of formalization and interaction thus seem well suited to an analysis of the basic but still complex processes through which early modern technology-related knowledge was generated, communicated, and put into practice.

CONCLUSION

The actual ways in which additional formalization and subsequent interaction of technology-related knowledge as sketched above might indeed have left their mark on technological developments in the early modern period have yet to be comprehensively discussed. In particular, with regard to the British industrialization process, there is an ongoing debate as to whether early modern transformations of technical knowledge represented a decisive factor for the onset of industrialization, or whether the major driving forces instead involved a matrix of diverse economic, social, cultural, and environmental factors, as Evans and Whitey have argued recently.[17] In the end, this question might quickly lead into the contested field of counterfactual history as one is more or less forced to ask, for example, how a particular technology or a particular project would have developed if certain types of knowledge had not been used by the artisans and other actors involved.

In any case, the early modern processes of formalization and interaction discussed here encompassed no unilateral move of the replacement of one knowledge regime—for matters of convenience, labeled artisanal knowledge—with another that was heavily based on knowledge shared in new representational media and institutions. It thus seems worthwhile to study in more detail the complex developments that *supplemented*, but definitely did not *replace*, established forms of knowledge that formalized mostly in the bodies of artisans and related to the employment of objects like tools and instruments. This, however, means to deliberately turn attention away from "science" as the main reference point for studying technology-related knowledge. From the perspective advocated here, the conventional concentration on the study of a presumed science/technology divide in the early modern period has overshadowed the complex developments of the formalization and subsequent interaction of technology-related knowledge. It deserves more attention not only in case studies, but also as part of a comprehensive history of early modern technology.

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[17] Chris Evans and Alun Withey, "An Enlightenment in Steel? Innovation in the Steel Trades of Eighteenth-Century Britain," *Technology and Culture*, 2012, 53:533–560.